

SPY 0DTE Gamma Explosion Bot

Phase 0 Strategy Document

Alpaca Markets · Paper Trading · Updated July 2026

This document reflects all live-session improvements through 2026-06-17. Changes since the May 2026 revision are: MAX_STRIKE_OFFSET cap on strike distance, ATR-adaptive SPY-level stop buffer, and June 1–17 live session records.

1. Strategy Overview

This bot targets same-day expiry (0DTE) call and put options on SPY (SPDR S&P; 500 ETF Trust). The core thesis is that out-of-the-money (OTM) options are effectively inert for most of the trading day until the underlying makes a strong directional move toward the strike. At that point, gamma acceleration causes the option to reprice explosively, often moving 5–20× in a matter of minutes. The bot identifies the moment before that explosion, enters cheaply, and exits at a pre-defined profit target or stop loss. It trades both calls (bullish) and puts (bearish), making it direction-agnostic.

Key Parameters

Parameter	Value / Detail
Underlying	SPY (S&P; 500 ETF, 0DTE options)
Broker / API	Alpaca Markets (alpaca-py SDK, paper trading)
ATR multiplier	$0.60 \times$ 5-day daily ATR — sets OTM strike offset
Max strike offset	\$4.50 hard cap — prevents elevated ATR from placing strikes beyond SPY's realized range
Strike window	Target ± 10 strikes (21 per side, \$0.50 increments)
Option sub timing	Deferred to 9:30 ET first RTH bar (avoids premarket price mismatch)
Re-subscription	Triggered when SPY moves $\pm \$3$ from last anchor; held symbol always pinned
Entry window	9:45 AM – 2:30 PM ET
Hard exit (time)	3:25 PM ET — force-close all positions
Strike zone filter	Activation only: SPY within 0.3% of strike ($\sim \$2.25$ at SPY \$750)
ATR5 gate	$\text{atr5} \geq 0.20$ — blocks entries when intrabar velocity too low for gamma move
Take profit	$1.50\times$ entry price (50% gain)
Peak trail activate	$1.20\times$ entry price — arms trailing stop once 20% gain is reached
Peak trail stop	88% of peak mid — exits if price falls 12% from highest point
Stop loss	$0.50\times$ entry price (50% loss)
SPY-level stop	Exit if SPY bar-close moves $\max(\$0.10, 0.75 \times \text{atr5_at_entry})$ against position direction
Max risk / trade	\$150 (position sized to this dollar amount at stop)
Daily loss limit	\$300 — bot stops new entries if breached
Trade cooldown	3 bars (3 min) between any two trades
Option price range	\$0.20 – \$10.00
Min consecutive bars	3 (green for calls, red for puts)

Note: SPX index options are not available on Alpaca as of June 2026. SPY is the equivalent — $\text{SPY price} \approx \text{SPX} / 10$, with identical 0DTE gamma dynamics.

2. Why This Works — The Options Physics

The Dead Zone

A deeply OTM 0DTE call option (e.g., strike \$5 above current SPY price) behaves like a lottery ticket. Delta is near zero (0.01–0.05), meaning a \$1 move in SPY changes the option price by only \$0.01–\$0.05. Theta (time decay) aggressively erodes value every minute. These two forces roughly cancel — the option chart looks flat for hours while SPY moves sideways.

The Activation Event

As SPY approaches the strike, the market begins pricing in real probability of expiring in-the-money. Delta surges from near-zero toward 0.30, 0.50, and beyond. Gamma — the rate of change of delta — is at its mathematical peak near the strike on expiry day. The transition from lottery ticket to real option can happen in 5–15 minutes. The edge: this transition is partially predictable. If SPY is in a strong sustained trend and approaching a pre-selected strike, the probability of an explosive option move is significantly elevated. The bot enters during the approach phase using momentum, proximity, and ATR velocity signals.

The Theta vs. Movement Race

For a 0DTE near-ATM option, theta does not wait. A typical option loses \$0.005–\$0.012 per minute to decay. Option price gain from SPY movement = $\text{delta} \times \Delta\text{SP}$ per minute. With $\text{delta} \approx 0.40$ in the activation zone:

atr5 (\$/min)	Est. SPY direction/min	Option gain/min	Theta loss/min	Net
0.45	~\$0.20	\$0.080	\$0.008	+\$0.072 strong
0.30	~\$0.13	\$0.052	\$0.008	+\$0.044 good
0.20	~\$0.09	\$0.036	\$0.008	+\$0.028 marginal
0.12	~\$0.05	\$0.020	\$0.010	+\$0.010 breakeven
0.10	~\$0.04	\$0.016	\$0.012	+\$0.004 theta wins

Below $\text{atr5} \approx 0.20$, theta wins the race almost every time. This is the basis of the ATR5 gate (Section 6).

3. Strike Selection — Dynamic, Per Bar

Strike selection is recomputed on every 1-minute SPY bar. The original pre-market fixed-strike approach was abandoned after a real failure: the bot started one morning with SPY at \$743 premarket, subscribed puts at \$735–\$736 — but the market opened at \$734. A \$9 gap meant wrong strikes all day. The fix: defer all option subscriptions until the first 9:30 ET RTH bar.

Startup Sequence

- Fetch last 5 completed daily bars via Alpaca REST → compute 5-day daily ATR baseline
- Pre-seed EMA5/EMA20 with last 30 RTH 1-min bars so indicators are warm at open
- VWAP resets to zero and accumulates from the 9:30 ET bar onward
- Wait for first 9:30 ET bar → use actual open price to compute strikes and subscribe

Strike Formula

The strike offset is capped at \$4.50 regardless of the 5-day ATR. This was added after two consecutive zero-trade days (Jun 12, Jun 15) when the 5-day ATR had escalated to ~\$15, producing a \$9 offset that placed strikes completely outside SPY's realized intraday range.

```
offset = min(0.60 × ATR_5day, 4.50)
call_target = round(SPY_open + offset, nearest $0.50)
put_target = round(SPY_open - offset, nearest $0.50)
```

Window: target ± 10 strikes = 21 calls + 21 puts = 42 OCC symbols

Example — Normal ATR regime (May 2026):

```
SPY open ~$737, ATR ~$5.20 → offset = min(0.60 × 5.20, 4.50) = min(3.12, 4.50) = $3.12
call_target = $740.00 | put_target = $734.00
```

Example — Elevated ATR regime (June 2026, with cap):

```
SPY open ~$753, ATR ~$15.0 → offset = min(0.60 × 15.0, 4.50) = min(9.00, 4.50) = $4.50
call_target = $757.50 | put_target = $748.50
Without cap: call_target = $762.00 (unreachable – SPY never got within $5 on Jun 12)
```

Mid-Session Re-Subscription

A background watcher checks SPY price every 10 seconds. If SPY has moved more than ±\$3.00 from the last subscription anchor, it re-subscribes a new window centred on the current price. If an open position exists, its symbol is always pinned. Old symbols stay subscribed harmlessly — the routing table filters non-target quotes.

4. Momentum Engine

Consumes SPY 1-minute bars from Alpaca StockDataStream (IEX free feed). Computes five indicators per bar and combines them into a directional signal. EMAs carry forward across sessions. VWAP resets at 9:30 ET daily. Pre-seeded with 30 historical RTH bars at startup so all indicators are meaningful from bar one.

Indicator	Formula	Purpose
EMA5	Exponential MA, period=5	Fast trend direction
EMA20	Exponential MA, period=20	Slow trend direction
VWAP	$\text{Sum}(\text{typical_price} \times \text{vol}) / \text{Sum}(\text{vol})$	Session anchor, resets 9:30 ET
ROC5	$(\text{close} - \text{close}[5]) / \text{close}[5]$	Rate of change over last 5 bars
atr5	Avg true range of last 5 one-min bars	Intrabar velocity — ATR5 gate
Consec bars	Count of consecutive green or red 1-min bars	Trend persistence filter

Signal Rules

Signal	Conditions — ALL must be true simultaneously	Enables
BULL	price > VWAP AND EMA5 > EMA20 AND ROC5 ≥ +0.03% AND consec green ≥ 3	Call entry evaluation
BEAR	price < VWAP AND EMA5 < EMA20 AND ROC5 ≤ -0.03% AND consec red ≥ 3	Put entry evaluation
NEUTRAL	Any condition above not met	No entries

Note: MIN_CONSEC_BARS was tightened from 2 to 3 after live testing showed that 2-bar signals produced too many false entries at the top or bottom of short-lived moves.

5. Proxy Delta — Solving the 0DTE Greeks Problem

Alpaca computes Greeks server-side using Black-Scholes. When $T = 0$ (expiry day), the formula is mathematically undefined. Alpaca returns null Greeks for all 0DTE contracts.

```
proxy_delta = (option_mid_now - option_mid_prev) / (SPY_price_now - SPY_price_prev)
```

Phase 0 Status: Proxy Delta Disabled

During live testing, SPY price updates only once per 1-minute bar while option quotes arrive many times per second. Between bar updates the SPY price is stale, so proxy delta stays at zero for most of the minute and would block all entries. For Phase 0, proxy delta is disabled (PROXY_DELTA_MIN = 0.0). Momentum, ATR5 gate, and strike proximity filters serve as the gatekeepers.

6. Entry Rules

All conditions below must be simultaneously true for a limit buy order to fire. Entry is triggered by incoming option quote ticks — not bar events — so the bot reacts within milliseconds of a quote update. An asyncio lock prevents concurrent evaluation.

#	Condition	Value / Detail
1	Time window	9:45 AM – 2:30 PM ET
2	ATR5 gate	$\text{atr5} \geq 0.20$ — 5-bar rolling ATR must show sufficient intrabar velocity. Blocks theta-bleed setups.
3	Momentum signal	BULL for calls BEAR for puts (3+ consecutive bars required)
4	Option price	\$0.20 – \$10.00
5	Directionality	Call: SPY below strike (OTM). Put: SPY above strike (OTM). Blocks ITM entries.
6	Strike proximity	Activation zone only: SPY within 0.3% of strike (~\$2.25 at SPY \$750)
7	Open positions	None — one position at a time in Phase 0
8	Trade cooldown	0 bars remaining (3-bar cooldown enforced after each close)
9	Daily loss gate	Not breached (\$300 cumulative daily loss limit)

Order Mechanics

- Limit buy at mid-price + 2% slippage buffer to improve fill probability
- Fill timeout: 30 seconds — order is cancelled if unfilled
- Bracket orders and sell limit orders are both rejected for Alpaca paper options. All exits use `close_position()`.

7. Exit Rules & Position Management

Exits are driven directly by the option quote stream. Every time a new quote arrives for the held symbol, `_evaluate_exit()` is scheduled as an asyncio task — sub-50ms response from quote arrival to order submission. A 30-second safety net provides fallback coverage. `_time_stop_watcher` force-closes at 3:25 PM ET.

Exit Trigger Priority

1. Update `peak_mid`: if current `mid` > `peak_mid`, store it

2. IF `mid` >= `entry` × 1.50 → exit reason='tp'

3. ELSE IF `mid` <= `entry` × 0.50 → exit reason='stop'

4. ELSE IF `peak_mid` >= `entry` × 1.20 (trail armed):
 `trail_stop` = `peak_mid` × 0.88
 IF `mid` <= `trail_stop` → exit reason='peak_trail'

5. ELSE: return — no action

Separately, at 3:25 PM ET: `_time_stop_watcher` → exit reason='time_stop'

Exit Type	Trigger Condition	Role
Take Profit	$mid \geq entry \times 1.50$ (50% gain)	Captures the full gamma explosion payoff. Checked first.
Hard Stop	$mid \leq entry \times 0.50$ (50% loss)	Caps maximum dollar loss. Wide threshold avoids false stops from spread noise.
Peak Trailing Stop	Arms at $peak_mid \geq entry \times 1.20$. Fires at $mid \leq peak \times 0.88$.	Captures partial recoveries. Requires genuine 20% move before arming.
Time Stop	Clock $\geq$ 3:25 PM ET	No 0DTE position survives to expiry.
SPY-Level Stop	$buf = \max(\$0.10, 0.75 \times atr5_at_entry)$ Call: $SPY\ bar_close < entry_SPY - buf$ Put: $SPY\ bar_close > entry_SPY + buf$	Trend-reversal signal using underlying. ATR-adaptive buffer reduces whipsaw false-stops on high-velocity sessions.

ATR-Adaptive SPY Stop — Detail

The SPY-level stop buffer was originally a fixed \$0.10. Analysis of 58 `spy_stop` exits across June sessions found that 27 of 52 exits (52%) saw SPY reverse within 3 bars — the position would have been profitable if held. \$0.10 is inside normal bar-to-bar SPY noise when `atr5` is 0.35–0.45. The fix: scale the buffer to the intrabar ATR at the time of entry.

atr5 at entry	Stop buffer	Context
0.20	\$0.15	Slightly wider than old fixed \$0.10 — calm session
0.30	\$0.225	Allows normal bar noise without stopping out
0.40	\$0.30	High-velocity session — real reversals are larger moves
0.50	\$0.375	Very volatile — genuine stop requires a significant adverse bar

First day live (June 17): 2 `spy_stops` out of 6 trades vs. 7–11/day in the prior week. Both that fired held 5 and 17 bars — actual reversals, not noise.

Why the Stop is 50%, Not 25%

0DTE options oscillate. For a \$0.76 option with a \$0.05 bid/ask spread, the mid can swing $\pm\$0.08$ (10%) on a flat SPY bar from quote fluctuation alone. A 25% stop would be triggered by normal spread noise. The 50% stop requires a genuine move against the position. Position sizing (`MAX_RISK_PER_TRADE` = \$150) keeps the absolute dollar loss bounded regardless of the wide percentage stop.

8. Risk Management

Position Sizing

Contracts are sized so that a full stop-loss hit equals exactly \$150. Dollar risk is consistent regardless of option price.

```
risk_per_contract = entry_price × (1 - 0.50) × 100 # 50% stop  
contracts = floor($150 / risk_per_contract)  
contracts = max(1, contracts) # always at least 1
```

Entry \$0.96 → risk/contract = \$48.00 → 3 contracts (\$144 at risk)

Entry \$0.41 → risk/contract = \$20.50 → 7 contracts (\$143.50 at risk)

Entry \$1.92 → risk/contract = \$96.00 → 1 contract (\$96 at risk)

Daily Loss Gate

- If cumulative realized P&L falls below −\$300, no new entries are taken
- Existing positions continue to be managed (TP / stop / trail / time-stop still active)
- Daily P&L is persisted to the dated trades CSV and restored on restart — restarting does not reset the gate

9. Live Sessions

Full Session Log (May 20 – June 17, 2026)

Days marked * include ghost-position cascade exits from the watchdog restart bug (Section 14). The 'Clean P&L' column excludes those anomalous exits and reflects actual strategy performance.

Date	Trades	Raw P&L	Clean P&L	Notes
May 20	5	+\$64	+\$64	First full session
May 26	4	\$-309	\$-309	Post-mortem session — 5 failure patterns identified
May 27	10	\$-159	\$-159	First day with fixes; trail threshold miscalibrated (1.05×)
May 29	5	+\$70	+\$70	First green day with ATR gate + trail at 1.20×
Jun 1	4	\$-124	\$-124	
Jun 2	2	\$-117	\$-117	
Jun 3 *	10	\$-148	+\$76	Ghost cascade: -\$224 excluded from clean
Jun 4 *	11	+\$1287	\$-58	Ghost cascade: +\$1,345 excluded from clean
Jun 5	18	+\$178	+\$178	Strong bear trend; 18 put trades
Jun 8	15	\$-324	\$-324	Elevated ATR; high spy_stop rate before adaptive fix
Jun 9	15	+\$265	+\$265	Strong bear leg; 4 TPs
Jun 10	14	\$-79	\$-79	
Jun 11	5	\$-1	\$-1	Flat
Jun 12	0	\$0	\$0	← Strike calibration failure (ATR=15.43, offset=\$9.26)
Jun 15	0	\$0	\$0	← Strike calibration failure (same root cause)
Jun 16	13	\$-19	\$-19	First day with MAX_STRIKE_OFFSET=\$4.50 cap
Jun 17	6	+\$82	+\$82	First day with ATR-adaptive SPY stop

Clean strategy total (May 20 – Jun 17): -\$121 | Raw total: +\$1,007 (almost entirely from Jun 4 ghost anomaly)

2026-05-29 — First Green Day with Full Filters — +\$70

First session with both ATR5 gate ($\text{atr5} \geq 0.20$) and correctly calibrated peak trail ($1.20\times / 88\%$). All five entries passed the ATR gate. Demonstrates the position management system converting partial moves from losses into gains.

#	Symbol	Side	Entry	Exit	atr5	Peak	Hold	Reason	P&L
1	C759	CALL	\$0.76	\$0.37	0.226	—	24m	stop	-\$117
2	P754	PUT	\$0.68	\$1.03	0.319	—	18m	tp	+\$140
3	P755	PUT	\$0.57	\$0.61	0.218	\$0.69 +21%	4m	peak_trail	+\$20
4	C759	CALL	\$0.31	\$0.34	0.243	\$0.38 +23%	5m	peak_trail	+\$27
5	P756	PUT	\$0.79	\$0.79	0.200	\$0.95 +20%	28m	peak_trail	+\$0

Trades 3 and 4 illustrate position management: both peaked 21–23% above entry, trail armed, exited +\$20 and +\$27. Without the trail, both hit hard stops at −\$28 and −\$45. The trail converted a combined −\$73 into +\$47 — a +\$120 swing on a +\$70 session.

2026-06-05 — Strong Bear Trend — +\$178

18 trades on a sustained bear day (SPY fell from ~\$750 to ~\$740). Demonstrates regime-aware operation: the bot tracked the declining trend through multiple re-entries as SPY stepped lower. High trade count reflects the bot correctly re-entering after short reversals that triggered `spy_stops`.

#	Symbol	Side	Entry	Exit	Hold	Reason	P&L
1	P749	PUT	\$1.10	\$1.155	2m	peak_trail	+\$11
2	P748	PUT	\$1.12	\$1.67	3m	tp	+\$110
3	C751	CALL	\$1.33	\$1.21	6m	spy_stop	-\$24
4	P749	PUT	\$1.54	\$1.21	4m	spy_stop	-\$33
5	C751	CALL	\$1.23	\$1.10	3m	spy_stop	-\$26
6	P749	PUT	\$1.16	\$1.25	5m	peak_trail	+\$18
7	P749	PUT	\$1.22	\$1.12	1m	spy_stop	-\$20
8	P748	PUT	\$1.09	\$1.16	8m	peak_trail	+\$14
9	P746	PUT	\$0.65	\$0.74	3m	peak_trail	+\$36
10	P745	PUT	\$0.67	\$0.75	2m	peak_trail	+\$32
11	P744	PUT	\$0.96	\$0.98	1m	peak_trail	+\$6
12	P744	PUT	\$1.15	\$1.15	1m	spy_stop	+\$0
13	P744	PUT	\$1.03	\$1.51	4m	tp	+\$96
14	P743	PUT	\$1.32	\$1.515	2m	peak_trail	+\$39
15	P742	PUT	\$1.00	\$0.81	1m	spy_stop	-\$57

1 6	P743	PUT	\$0.87	\$0.76	1m	spy_stop	-\$33
1 7	P742	PUT	\$0.44	\$0.51	1m	peak_trail	+\$42
1 8	P740	PUT	\$0.91	\$0.80	1m	spy_stop	-\$33

Session total: +\$178 | 18 trades | 2 TP / 11 peak_trail / 5 spy_stop. Note the high spy_stop rate (5 of 18) — this is the pre-adaptive-fix regime. With the Jun 17 adaptive buffer, these would likely be reduced.

2026-06-09 — Strong Bear Leg — +\$265

Best clean session to date. SPY dropped ~\$15 intraday from ~\$747 to ~\$722. Four take-profit exits across multiple put strikes as the bot tracked the move lower. Demonstrates the strategy at its best: sustained directional trend, high atr5, multiple clean TP hits.

#	Symbol	Side	Entry	Exit	atr5	Hold	Reason	P&L
1	C748	CALL	\$0.66	\$0.76	0.31	1m	peak_trail	+\$40
2	P743	PUT	\$1.08	\$0.95	0.28	1m	spy_stop	-\$26
3	P742	PUT	\$1.46	\$2.16	0.35	2m	tp	+\$140
4	P740	PUT	\$1.72	\$2.16	0.41	2m	peak_trail	+\$44
5	P738	PUT	\$2.24	\$2.52	0.38	1m	peak_trail	+\$28
6	P735	PUT	\$2.65	\$2.09	0.44	2m	spy_stop	-\$56
7	P735	PUT	\$1.51	\$1.87	0.32	6m	peak_trail	+\$36
8	P732	PUT	\$1.67	\$2.45	0.39	2m	tp	+\$78
9	P729	PUT	\$1.86	\$1.46	0.42	1m	spy_stop	-\$40
10	P731	PUT	\$1.97	\$1.79	0.38	1m	spy_stop	-\$18
11	P731	PUT	\$2.27	\$3.44	0.45	2m	tp	+\$117
12	P726	PUT	\$2.03	\$3.05	0.43	3m	tp	+\$102
13	P722	PUT	\$2.72	\$2.11	0.37	3m	spy_stop	-\$61
14	P730	PUT	\$1.50	\$1.16	0.29	1m	spy_stop	-\$68
15	C737	CALL	\$0.82	\$0.65	0.25	1m	spy_stop	-\$51

Session total: +\$265 | 15 trades | 4 TP / 5 peak_trail / 6 spy_stop. 4 take-profit exits on a single trending session — the strategy working as designed. The 6 spy_stops are notable: all occurred in the same pre-adaptive-fix period, with the fixed \$0.10 buffer insufficient for the 0.35–0.45 atr5 regime.

2026-06-17 — First Day with Adaptive SPY Stop — +\$82

First session with both MAX_STRIKE_OFFSET cap and ATR-adaptive SPY stop live simultaneously. Only 2 spy_stops out of 6 trades (vs. 7–11/day in the prior week). Both spy_stops held for 5 and 17 bars before exiting — genuine reversals, not noise. SPY was range-bound ~\$750, alternating between bull and bear momentum.

#	Symbol	Side	Entry	Exit	atr5 @ entry	buf	Hold	Reason	P&L
1	C752	CALL	\$1.92	\$1.81	0.31	\$0.23	5 bars	spy_stop	-\$11
2	P750	PUT	\$1.91	\$2.30	0.33	—	3 bars	peak_trail	+\$39
3	C752	CALL	\$1.90	\$1.99	0.29	—	10 bars	peak_trail	+\$9
4	P750	PUT	\$1.93	\$2.46	0.34	—	8 bars	peak_trail	+\$53
5	P749	PUT	\$2.26	\$2.00	0.29	\$0.22	17 bars	spy_stop	-\$26
6	C750	CALL	\$2.19	\$2.37	0.32	—	25 bars	peak_trail	+\$18

Session total: +\$82 | 6 trades | 4W / 2L | 4/6 peak_trail | 2/6 spy_stop. Trade 1: $\text{buf} = \max(\$0.10, 0.75 \times 0.31) = \0.23 — SPY closed \$0.23+ against entry, genuine reversal. Trade 5: $\text{buf} = \max(\$0.10, 0.75 \times 0.29) = \0.22 — held 17 bars, confirming the reversal was real. With the old fixed \$0.10 buffer, both would have stopped out on bar 1.

2026-06-12 and 2026-06-15 — Strike Calibration Failure

Both sessions had zero trades. The bot ran all day, processed 390 bars each session, detected momentum signals — and never entered. Root cause: the 5-day daily ATR had escalated to \$13.69–\$15.43 during the May–June volatility shift.

Without the offset cap, this produced:

```
startup_atr=15.43 → offset = 0.60 × 15.43 = $9.26  
call_target = $762.50 | put_target = $744.00  
SPY ranged $751–$757 all day – $5+ outside the approach zone (0.7% = $5.27 at $753)
```

The bot was technically functioning correctly. The strike selection was producing unreachable targets with no warning. Fix: MAX_STRIKE_OFFSET = \$4.50 cap, deployed June 16. Jun 16 had 13 trades.

10. Terminal Interface & Bot Controls

The bot prints a status block to the terminal. With a position open it updates every 5 seconds; when flat, once per minute (aligned to bar events).

Status Block — Position Open

```
-----
12:16:44 ET | SPY $749.85 | v BEAR | VWAP $750.63
EMA5 $749.99 EMA20 $750.29 | ROC -0.0004 | consec -4
.....
POSITION: SPY260526P00749000 (PUT $749) | in trade 0m 12s
Entry $0.84 × 3 contracts | Entry SPY $750.45 SPY stop ↑$750.69
Mid $0.97 (+15%) | TP $1.26 Stop $0.42 | Trail ARMED @ $1.008
Unrealised P&L: +$39.00 | min +$0.00 max +$39.00
.....
Daily P&L -$144.00 | Trades 2 | Gate open
-----
```

The SPY stop level is now displayed as an adaptive value (entry_SPY ± buf where buf = max(\$0.10, 0.75 × atr5_at_entry)), not a fixed \$0.10 offset.

Keyboard Commands

Key	Action
q + Enter	Quit cleanly — cancels open orders, closes all positions, exits process
r + Enter	Restart — cancels unfilled orders, leaves open positions on Alpaca, relaunches via os.exec(). Position recovered from REST on next startup.
t + Enter	Print today's trade list — symbol, entry/exit, hold time, reason, P&L per trade, session total

11. Technical Architecture

Single Python process using asyncio for concurrency. Three WebSocket streams run as concurrent tasks. All state is in-memory; persistence is via dated CSV files and the Alpaca REST API.

Component	Technology	Notes
Language	Python 3.11+	
SDK	alpaca-py $\geq 0.43.0$	Older alpaca-trade-api has no options support
SPY bars	StockDataStream (IEX)	1-min bars, free tier
Option quotes	OptionDataStream	Real-time indicative quotes, msgpack format
Order events	TradingStream	Real-time fill/cancel events
Entry orders	TradingClient.submit_order()	Limit buy, BUY_TO_OPEN intent
Exit orders	TradingClient.close_position()	Only working exit for options on Alpaca paper
Trade log	logs/trades_YYYY-MM-DD.csv	Per-date, append-only; restored on restart
Session log	logs/bot_YYYY-MM-DD.log	Per-date structured INFO log; all signals, fills

Asyncio Task Graph

```

asyncio.gather(
    feed.start() # 3 WebSocket streams (stock + options + trading)
    _option_subscriber() # waits for 9:30 bar → subscribes option symbols
    _resubscribe_watcher() # re-subscribes if SPY moves  $\pm \$3$ 
    _exit_monitor() # 30-second safety net (WebSocket reconnect gaps)
    _status_loop() # terminal display
    _time_stop_watcher() # force-close at 3:25 ET
)

# Quote-driven exits (not a gathered task):
# on_option_quote() → asyncio.create_task(_evaluate_exit(quote))
# Sub-50ms from quote arrival to order submission.
```


12. Operating Procedure

Step	Time (ET)	Action
1	Any time	python main.py — fetches ATR, pre-seeds EMAs, connects streams
2	Pre-market	Streams connected; momentum engine warms up (VWAP accumulation deferred to 9:30)
3	9:30 (first bar)	Market open event fires → options subscribed at actual open price
4	9:45	Entry window opens — bot is fully autonomous
5	9:45 – 2:30	Active session — type 't' for trades, status auto-prints every 5s (60s when flat)
6	3:25	Time stop — all positions force-closed via close_position()
7	After close	Review logs/trades_YYYY-MM-DD.csv for session P&L and signal analysis

The bot can be started the night before or any time pre-market. Option subscription is always deferred to the 9:30 ET bar, so premarket SPY price movements never affect strike selection. On restart mid-session ('r' key): any existing Alpaca position is recovered automatically, and daily P&L is restored from the dated trades CSV so the loss gate remains accurate.

13. Phase 0 Success Criteria

Paper trading must run for a minimum of 15 trading days before live deployment is considered. All criteria below must be satisfied:

Criterion	Target	Rationale
Trading days	≥ 15	Statistically meaningful sample across market regimes
Total trades	≥ 20	Enough fills to evaluate signal quality
Win rate	> 25%	Profitable at 30%+ given 1.5× payoff ratio
Avg winner / loser	> 1.5×	Minimum edge for positive expectancy at 33% win rate
Daily loss limit	100% respected	No session must breach the \$300 gate
Order fill rate	> 90% / 30s	Confirms options are liquid enough to trade

Current status (as of Jul 2): 20+ active trading days, 220+ trades. The ghost-position cascade is resolved (see Section 15). The duplicate-order race must be fully closed (Fix #2, planned) before advancing to live trading.

14. Known Limitations & Roadmap

Current Limitations (Phase 0)

- No SPX — Alpaca does not support index options as of June 2026
- Proxy delta disabled — SPY price only updates once per 1-min bar; delta stays zero between bars
- One position at a time — misses concurrent call and put opportunities
- Exit fills at the bid (market order) — effective TP slightly below 1.50×; cost \$0.01–\$0.03/share
- Duplicate-order race partially open — cancelled buys can still double-fill; sweeper contains it (see Section 15)

Improvements Implemented in Live Testing

Improvement	Date	Impact
Routing table bug fix: each OCC symbol parses its own strike	May 22	Fixed incorrect zone checks and ITM entries across alt symbols
ITM directionality guard (call blocked if $SPY \geq \text{strike}$)	May 22	Prevents entering after gamma explosion has already occurred
Activation-zone-only entries (removed approach zone)	May 22	Reduces reversals from entries made too far from the strike
ATR5 gate: $\text{atr5} \geq 0.20$ required at entry	May 27	Blocks theta-bleed setups on low-velocity sessions
Peak trailing stop (arms at 1.20×, trails at 88% of peak)	May 27	Captures partial recoveries without triggering on spread noise
Price-based option re-subscription ($\pm \$3$ SPY move)	May 22	Keeps subscribed strikes relevant as SPY drifts intraday
Position recovery on restart via Alpaca REST	May 22	Restart does not abandon open positions
Daily P&L restoration from dated CSV on restart	May 22	Loss gate remains accurate across restarts
$\text{MAX_STRIKE_OFFSET} = \4.50 cap on strike distance	Jun 16	Prevents elevated 5-day ATR from placing strikes beyond SPY's realized range
ATR-adaptive SPY stop: $\text{buf} = \max(\$0.10, 0.75 \times \text{atr5_at_entry})$	Jun 17	Reduces whipsaw false-stops from 52% → ~33% on first confirmed session
Cancelled-buy fill recovery in <code>buy_limit()</code> (duplicate-race Fix #1)	Jun 29	Recovers fills that land despite a cancelled wait — partial fix, see Section 15
EOD clean exit: confirm flat via REST, then <code>os._exit(0)</code>	Jun 30	Eliminates the watchdog restart cascade — verified Jul 1 & Jul 2
ORB clock-hour shadow filter (observe-only)	Jun 30	Logs would-be blocks per entry; live validation before gating — see Section 15

RESOLVED — Ghost Position / Watchdog Restart Cascade (Fixed Jun 30)

The bot has a watchdog that restarts the process if it goes silent for >20 seconds. Under the end-of-day sequence this produced a cascade of erroneous exits:

- Time stop fires at 3:25 PM — bot closes positions, then tears down the WebSocket feeds
- The feed teardown blocks the event loop for >20 seconds (even with zero open positions)
- Watchdog fires, kills the process, restarts it
- The fresh instance boots past 3:25 PM — immediately re-fires the time stop, hangs again

- Loop continues until the watchdog window closes at 3:35 PM (observed 15 restarts on Jun 30)

Confirmed on Jun 3 (–\$224 anomaly), Jun 4 (+\$1,345 anomaly — an untracked position happened to have a wide ask at 3:25 PM), and Jun 30 (15-restart loop, no positions affected). Key diagnostic: the Jun 25/29/30 sessions hung with ZERO open positions, proving the hang is in the feed teardown itself, not the `close_position()` fill-wait.

Fix (deployed Jun 30): the time stop no longer tears down the feeds. After force-closing, it confirms the account is flat via Alpaca REST (retrying stubborn closes up to 8×), then hard-exits with `os._exit(0)`. The watchdog is a daemon thread, so it dies with the process — no restart, no cascade. Verified live Jul 1 and Jul 2: flat-check plus exit completed in under 100 ms both days.

Planned for Phase 1

Improvement	Priority	Rationale
Duplicate-race Fix #2: keep <code>entry_pending</code> latched on ambiguous buy outcome until reconciled	Critical	Fix #1 loses the status-check race — 3 ghost events in 2 days (Jul 1–2) prove the gap
ORB filter go/no-go decision after shadow-data review	High	June backtest +\$229 over 19 sessions; first live shadow day +\$70 — needs more reversal days
VWAP extension cap: block entries when $ \text{SPY} - \text{VWAP} > \1.25	High	May 26 Trades 3 & 4 entered \$1.32–\$1.77 below VWAP — high mean-reversion risk
Failed-strike session cooldown: block same strike for 60 min after a stop	High	P748 was re-entered 23 min after stopping out on May 26 and May 27
Extended post-TP cooldown (10 bars vs current 3)	Medium	May 26 Trade 3 chased an exhausted move 3.5 min after Trade 2 TP
Tick-level SPY price from <code>OptionDataStream</code> → re-enable proxy delta	Medium	Directional confirmation without waiting for the next 1-min bar
PEAK_TRAIL_PCT tightening from 0.88 to 0.92–0.93	Medium	Recovers ~\$11 on sessions where the trail gives back too much from peak
IV rank filter (prior-day IV as baseline)	Low	Avoid overpriced options on high-IV days
Multiple concurrent positions with independent per-position risk budgets	Low	Misses simultaneous call and put gamma setups

15. July 2026 Update — Fixes Verified Live, ORB Shadow Filter, Public Dashboard

Session Results: Jun 30 – Jul 2 (three consecutive green days)

Date	Trades	Booked P&L	Ghost P&L	Actual P&L	Notes
Jun 30	3	+\$148	—	+\$148	3/3 winners; flat afternoon correctly sat out
Jul 1	7	+\$143	+\$27	+\$170	1 duplicate-race ghost, swept in 9 min
Jul 2	18	+\$207	\$-18	+\$189	Reversal day; 2 ghosts swept in 1–2 min

Duplicate-Order Race — Fix #1 Deployed, Gap Confirmed

Root cause: `on_option_quote()` awaits `_evaluate_entry()` inline in the stream callback, so a stream teardown propagates `asyncio.CancelledError` into the buy fill-wait. The finally-clause cleared `entry_pending` unconditionally, letting the next quote tick (measured 2–6 ms later) submit a second buy while the first order was still live on Alpaca's books.

Fix #1 (Jun 29): the `CancelledError` branch in `buy_limit()` re-checks the order status and, if filled, returns the order as a normal fill instead of re-raising. Live result: the recovery never fired — in all three observed events (Jul 1: one, Jul 2: two) the status check ran before Alpaca registered the fill, so the duplicate still occurred. The ghost sweeper contained every event within 1–9 minutes (net ghost P&L this week: +\$9). Fix #2 — keeping `entry_pending` latched on any ambiguous outcome until reconciled against REST — is the planned complete fix.

ORB Clock-Hour Shadow Filter (observe-only since Jun 30)

Concept: record the first 1-min bar of each clock hour (ORH/ORL). If the current hour's first bar printed BOTH a higher high and higher low than the prior hour's → calls only. Both lower → puts only. Mixed → no restriction. The filter runs in shadow mode: it logs BLOCK/ALLOW on every real entry signal but never prevents the trade.

June backtest (19 sessions, 189 trades): raw month P&L +\$643 → +\$872 with the filter, a +\$229 improvement from blocking 24 losses (–\$1,052) against 25 wins (+\$823). On a clean basis excluding the Jun 3/4 cascade artifacts: –\$478 → –\$249 — the filter cuts the clean-month loss nearly in half. The effect is broad (helped 11 of 19 days), not one lucky session.

First live shadow signal — Jul 2 (reversal day): 5 BLOCKs logged. Blocked losses \$208 (two morning call burners into the falling open, one counter-trend put), blocked wins \$138. Net effect +\$70 — the day would have been +\$277 filtered vs +\$207 actual. Jun 30 and Jul 1 (aligned trend days): zero blocks, all winners allowed through — the filter does not interfere with good days.

Caveat: the live filter saw a pre-market bar as the 8:00 hour reference on Jul 2, which set the opening put bias — the backtest did not model pre-market data identically. Worked in our favor here; monitoring across more sessions before the go-live decision.

Public KPI Dashboard

A rolling 30-calendar-day KPI dashboard is now auto-published daily at 15:45 ET to

milgar7969.github.io/spy-0dte-dashboard — equity curve, daily P&L, win rate, profit factor, breakdowns by exit reason / side / entry hour, ghost-adjusted actual P&L, and a live ORB shadow-filter scoreboard. Generated from the bot's own CSV and log files (`kpi_dashboard.py`); restart-recovery force-close artifacts (`time_stop` exits held under 60 s) are excluded from all KPIs.

SPY 0DTE Gamma Explosion Bot · Phase 0 · Updated July 2026